MTH2008 – Real Analysis

2026/27

Theorem

Suppose that f is continuous on a finite closed interval $[a, b]$. If

$$\alpha = \inf_{a \le x \le b} f(x) \quad \text{and} \quad \beta = \sup_{a \le x \le b} f(x),$$

then α and β are respectively the minimum and maximum of f on $[a, b]$.

Namely, there exist $x_1, x_2 \in [a, b]$ with

$$f(x_1) = \alpha \quad \text{and} \quad f(x_2) = \beta.$$

Theorem (Intermediate Value Theorem)

Suppose that f is continuous on $[a, b]$, $f(a) \neq f(b)$, and μ is between $f(a)$ and $f(b)$. Then $f(c) = \mu$ for some c in (a, b).

Uniform continuity

Definition

A function f is **uniformly continuous** on a subset S of its domain if, for every $\epsilon > 0$, there is a $\delta > 0$ such that, for all $x, x' \in S$ with $|x - x'| < \delta$

$$|f(x) - f(x')| < \epsilon.$$

We emphasise that in this definition δ depends only on ϵ and S and not on the particular choice of x and x', provided that they are both in S.

Example

200
180
160
140
120
100
80
60
40
mm
40
60
80
100
120
140
160
180
200
220
240
260
280

Theorem

If f is continuous on a closed and bounded interval $[a, b]$, then f is uniformly continuous on $[a, b]$.

Corollary

If f is continuous on a set T, then f is uniformly continuous on any finite closed interval contained in T.

More about monotonic functions

Observe that if f is monotonic on an interval I, then f is either continuous or has a jump discontinuity at each $x \in I$. This and the intermediate value Theorem provide the key to the proof of the following theorem.

Theorem

Let $a, b \in \mathbb{R}$ with $a < b$. If f is real-valued, monotonic and non-constant on $[a, b]$, then f is continuous on $[a, b]$ if and only if its range $R_f = \{f(x) \mid x \in [a, b]\}$ is the closed interval whose boundary is the set $\{f(a), f(b)\}$.

Theorem Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that f is real valued, strictly increasing and continuous on $[a, b]$, and let $f(a) = c$ and $f(b) = d$. There exists a unique function $g : [c, d] \to \mathbb{R}$, such that, for all $x \in [a, b]$ and all $y \in [c, d]$,

$$g(f(x)) = x \quad \text{and} \quad f(g(y)) = y.$$

Moreover, g is continuous and strictly increasing on $[c, d]$.

2.3 Differentiable functions (single variable)

In calculus we study differentiation, emphasising rules for calculating derivatives. Here we consider the theoretical properties of differentiable functions, defining the derivative of a function in terms of limits.

Definition

A function f is **differentiable** at an interior point x_0 of its domain, if the quotient of differences, with $x \neq x_0$,

$$\frac{f(x) - f(x_0)}{x - x_0},$$

approaches a limit as x approaches x_0, in which case the limit is called the **derivative of f at x_0**, and is denoted by $f'(x_0)$. In other words,

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

It is sometimes convenient to let $x = x_0 + h$ and write the above equation as

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

If f is defined on an open set S, we say that f is **differentiable on S** if f is differentiable at every point of S. Observe that, if f is differentiable on S, then f' is a function on S.

- We say that f is **continuously differentiable** on S if f' is continuous on S.
- If f is differentiable on a neighbourhood of x_0, it is reasonable to ask if f' is differentiable at x_0. If so, we denote the derivative of f' at x_0 by $f''(x_0)$. This is the **second derivative of f at x_0**, and it is also denoted by $f^{(2)}(x_0)$.
- Continuing inductively, if $f^{(n-1)}$ is defined on a neighbourhood of x_0, then the **n-th derivative of f at x_0**, denoted by $f^{(n)}(x_0)$, is the derivative of $f^{(n-1)}$ at x_0.
- For convenience we define the **zero-th derivative** of f to be f itself, namely $f^{(0)} = f$.

Example

Interpretations of the derivative

- If $f(x)$ is the position of a particle at time $x \neq x_0$, the quotient of differences

$$\frac{f(x) - f(x_0)}{x - x_0}$$

is the average velocity of the particle between times x_0 and x. As x approaches x_0, the average applies to shorter and shorter intervals. Therefore, it makes sense to regard the limit

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0},$$

if it exists, as the particle's **instantaneous velocity at time** x_0.

This interpretation may be useful even if x is not time, in which case, we regard $f'(x_0)$ as the **instantaneous rate of change of** $f(x)$ **at** x_0, regardless of the specific nature of the variable x.

- The derivative also has a geometric interpretation. The equation of the line through two points $(x_0, f(x_0))$ and $(x_1, f(x_1))$ on the curve $y = f(x)$ is

$$y = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0).$$

Varying x_1 generates lines through $(x_0, f(x_0))$ that rotate into the line

$$y = f(x_0) + f'(x_0)(x - x_0)$$

as x_1 approaches x_0. This is the **tangent** to the curve $y = f(x)$ at the point $(x_0, f(x_0))$.

Properties of differentiable functions

Theorem

If f is differentiable at x_0, then f is continuous at x_0.

WARNING: The converse of this theorem is false, since a function may be continuous at a point without being differentiable at the point.

Theorem

If f and g are differentiable at x_0, then $f+g$, $f-g$, and $f \cdot g$ are differentiable at x_0, with

- $(f+g)'(x_0) = f'(x_0) + g'(x_0)$;
- $(f-g)'(x_0) = f'(x_0) - g(x_0)$;
- $(f \cdot g)'(x_0) = f'(x_0) \cdot g(x_0) + f(x_0) \cdot g'(x_0)$.

The quotient f/g is differentiable at x_0 if $g(x_0) \neq 0$, with

- $$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0) \cdot g(x_0) - f(x_0) \cdot g'(x_0)}{(g(x_0))^2}.$$

Theorem (Chain rule)

Suppose that g is differentiable at x_0 and f is differentiable at $g(x_0)$. The composite function $h = f \circ g$, defined by

$$h(x) = f(g(x)),$$

is differentiable at x_0, with

$$h'(x_0) = f'(g(x_0))g'(x_0).$$

200
180
160
140
120
100
80
60
40
mm
40
60
80
100
120
140
160
180
200
220
240
260
280

Definition

- If f is a function defined on an interval $[x_0, b)$, the **right-hand derivative of f at x_0** is defined to be

$$f'_+(x_0) = \lim_{x \to x_0+} \frac{f(x) - f(x_0)}{x - x_0}$$

  if the limit exists.

- If f is a function is defined on an interval $(a, x_0]$, the **left-hand derivative of f at x_0** is defined to be

$$f'_-(x_0) = \lim_{x \to x_0-} \frac{f(x) - f(x_0)}{x - x_0}$$

  if the limit exists.

- We say that f is **differentiable on the closed interval** $[a, b]$ if f is differentiable on the open interval (a, b) and $f'_+(a)$ and $f'_-(b)$ both exist.

- We say that f is **continuously differentiable on** $[a, b]$ if f is differentiable on $[a, b]$, f' is continuous on (a, b) with

$$f'_+(a) = \lim_{x \to a^+} f'(x) \quad \text{and} \quad f'_-(b) = \lim_{x \to b^-} f'(x).$$

Extreme Values

Definition

Let f denote a real-valued function. We say that $f(x_0)$ is a **local extreme value** of f if there is a $\delta > 0$ such that $f(x) - f(x_0)$ does not change sign on $(x_0 - \delta, x_0 + \delta) \cap D_f$.

- $f(x_0)$ is a **local maximum value** of f if

$$f(x) \leq f(x_0)$$

for all $x \in (x_0 - \delta, x_0 + \delta) \cap D_f$. In which case, the point x_0 is called a **local extreme point** of f, or, more specifically, a **local maximum point** of f.

- $f(x_0)$ is a **local minimum value** of f if

$$f(x) \geq f(x_0)$$

for all $x \in (x_0 - \delta, x_0 + \delta) \cap D_f$. In which case, the point x_0 is called a **local extreme point** of f, or, more specifically, a **local minimum point** of f.

Theorem

If f is differentiable at a local extreme point $x_0 \in D_f^0$, then $f'(x_0) = 0$.

If $f'(x_0) = 0$, we say that x_0 is a **critical point** of f. The above Theorem implies that every local extreme point of f at which f is differentiable is a critical point of f. The converse is false. For example, 0 is a critical point of $f(x) = x^3$, but not a local extreme point.

Rolle's Theorem

Theorem (Rolle's Theorem)

Let $a, b \in \mathbb{R}$ with $a < b$ be fixed. Suppose that f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), and $f(a) = f(b)$. Then $f'(c) = 0$ for some c in the open interval (a, b).

Theorem 2.3.5 (Intermediate value theorem for derivatives)

Let $a, b \in \mathbb{R}$ with $a < b$ be fixed. Suppose that f is differentiable on $[a, b]$, that $f'_+(a) \neq f'_-(b)$, and that μ is between $f'_+(a)$ and $f'_-(b)$. Then $f'(c) = \mu$ for some c in (a, b).

Theorem (Generalised mean value theorem)

Let $a, b \in \mathbb{R}$ with $a < b$ be fixed. If f and g are continuous functions on the closed interval $[a, b]$ and differentiable on the open interval (a, b), then

$$(g(b) - g(a))f'(c) = (f(b) - f(a))g'(c)$$

for some c in (a, b).

Theorem (Mean value theorem)

Let $a, b \in \mathbb{R}$ with $a < b$ be fixed. If f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), then

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

for some c in (a, b).

Consequences of the mean value theorem

Theorem

If $f'(x) = 0$ for all x in (a, b), then f is constant on (a, b).

Theorem

If f' exists and does not change sign on (a, b), then f is monotonic on (a, b).

Theorem

If $|f'(x)| \leq M$ for all $x \in (a, b)$, then for all $x, x' \in (a, b)$,

$$|f(x) - f(x')| \leq M|x - x'|.$$

3 – Infinite Sequences and Series

Sequences and series are the workhorse of analysis and forms the foundations of most mathematical applications they appear, for instance, in signal analysis (Fourier and Taylor series) and physics (general harmonic motion).

An **infinite sequence** (more briefly, a **sequence**), of real numbers is a real-valued function defined on a set of integers $\{n \in \mathbb{Z} \mid n \geq k\}$, for some $k \in \mathbb{Z}$. We call the values of the function the **terms** of the sequence. We denote a sequence by listing its terms in order; thus,

$$(s_n)_{n=k}^{\infty} = (s_k, s_{k+1}, \dots).$$

When $k = 1$, we may sometimes write $(s_n)_{n\in\mathbb{N}}$ for $(s_n)_{n=1}^{\infty}$.

Important note: A sequence and a set are different. Indeed, a set may not be ordered, and a set does not contain repeated elements.

To highlight this, unlike in the lecture notes, we will use the notation $(s_n)_{n=k}^{\infty}$ over the notation $\{s_n\}_{n=k}^{\infty}$ – However, you may use which one you find most natural.

Example

Given a sequence $(s_n)_{n=k}^{\infty} = (s_k, s_{k+1}, \dots)$, as above, for $n \in \mathbb{N}$, the real number s_n is the $(n-k+1)$**-th term** of the sequence.

Note, two sequences comprising of the same terms but with an index shift are considered to be identical, for instance

$$\left(\frac{1}{n-2}\right)_{n=3}^{\infty} \quad \text{and} \quad \left(\frac{1}{n}\right)_{n=1}^{\infty}$$

are considered to be identical.

The interesting questions about a sequence $(s_n)_{n=k}^{\infty}$ of real numbers concern the behaviour of s_n for large n.

Definition (Convergent sequences)

A sequence $(s_n)_{n=k}^{\infty}$ of real numbers **converges to a limit** s if for every $\epsilon > 0$ there exists an integer N such that, for all integers $n > N$,

$$|s_n - s| < \epsilon$$

In this case, we say that $(s_n)_{n=k}^{\infty}$ is **convergent** and write

$$\lim_{n\to\infty} s_n = s.$$

A sequence that does not converge is called **divergent**.

Example


200
180
160
140
120
100
80
60
40
mm
40
60
80
100
120
140
160
180
200
220
240
260
280

We say that the terms of a sequence $(s_n)_{n=k}^{\infty}$ satisfy a given condition for all n if s_n satisfies the condition for all integers $n \geq k$, or for large n if there is an integer $N > k$ such that s_n satisfies the condition whenever $n \geq N$.

For example, the terms of the sequnce $(1/n)_{n=1}^{\infty}$ are positive for all n, while those of $(1 - 7/n)_{n=1}^{\infty}$ are positive for $n > N = 8$.

Definition (Sequences diverging to infinity)

Let $(s_n)_{n=1}^{\infty}$ denote a sequence of real numbers. We write

$$\lim_{n\to\infty} s_n = \infty$$

if for any real number a, there exists an $N \in \mathbb{N}$, such that $s_n > a$ for all integers $n \geq N$, in which case we say that $(s_n)_{n=k}^{\infty}$ **diverges to** ∞. Similarly, we write

$$\lim_{n\to\infty} s_n = -\infty$$

if for any real number a, there exists an $N \in \mathbb{N}$, such that $s_n < a$ for all integers $n \geq N$, in which case we say that $(s_n)_{n=k}^{\infty}$ **diverges to** $-\infty$.

Example

Our next result enables us to apply the theory of limits we have developed for function to some sequences.

Theorem

Let $\lim_{x\to\infty} f(x) = L$, where L is in the extended reals, and suppose that $s_n = f(n)$ for $n \in \mathbb{N}$ large. Then

$$\lim_{n\to\infty} s_n = L.$$

Example

200
180
160
140
120
100
80
60
40
mm
40
60
80
100
120
140
160
180
200
220
240
260
280

Definition (Subsequence)

A sequence $(t_m)_{m=l}^{\infty}$ is a **subsequence** of a sequence $(s_n)_{n=k}^{\infty}$ if

$$t_m = s_{n_m}$$

where $(n_m)_{m=l}^{\infty}$ is a strictly increasing infinite sequence of integers, with $n_l \geq k$.

We often write the subsequence $(t_k)_{k=l}^{\infty}$ by $(s_{n_m})_{m=l}^{\infty}$.

Note that $(s_n)_{n=k}^{\infty}$ is a subsequence of itself, as can be seen by taking $n_m = m$. All other subsequences of $(s_n)_{n=k}^{\infty}$ are obtained by deleting terms from $(s_n)_{n=k}^{\infty}$ and leaving those remaining in their original relative order.

Example

200
180
160
140
120
100
80
60
40
mm
40
60
80
100
120
140
160
180
200
220
240
260
280

Theorem

Let $(s_n)_{n=1}^{\infty}$ denote a sequence of real numbers. If there exists an $s \in \mathbb{R}$, with

$$\lim_{n\to\infty} s_n = s,$$

then

$$\lim_{m\to\infty} s_{n_m} = s$$

for every subsequence $(s_{n_m})_{m=l}^{\infty}$ of $(s_n)_{n=1}^{\infty}$.

For a given set S we defined **limit points** in terms of neighborhoods: $\overline{x}$ is a limit point of a set S if every neighborhood of $\overline{x}$ contains points of S distinct from $\overline{x}$. Our next result shows that an equivalent definition can be stated in terms of sequences.

Theorem

A point $\overline{x}$ is a limit point of a set S if and only if there exists a sequence $(x_n)_{n\in\mathbb{N}}$ of points in S such that $x_n \neq \overline{x}$ for all $n \in \mathbb{N}$, and

$$\lim_{n\to\infty} x_n = \overline{x}.$$

Theorem

(a) If $(x_n)_{\mathbb{N}}$ is bounded sequence of real numbers, then $(x_n)_{n\in\mathbb{N}}$ has a convergent subsequence.

(b) If $(x_n)_{n\in\mathbb{N}}$ is a sequence of real numbers unbounded above, then $(x_n)_{n\in\mathbb{N}}$ has a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ such that

$$\lim_{k\to\infty} x_{n_k} = \infty.$$

(c) If $(x_n)_{n\in\mathbb{N}}$ is a sequence of real numbers unbounded below, then $(x_n)_{n\in\mathbb{N}}$ has a subsequence $(x_{n_k})_{k\in\mathbb{N}}$ such that

$$\lim_{k\to\infty} x_{n_k} = -\infty.$$

200
180
160
140
120
100
80
60
40
mm
40
60
80
100
120
140
160
180
200
220
240
260
280

200
180
160
140
120
100
80
60
40
mm
40
60
80
100
120
140
160
180
200
220
240
260
280